Project: Worked examples
Title: Voltage drop and I²R loss
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

## 1. Scope and method

Voltage drop along a circuit in volts and per cent of the nominal voltage, the conductor power loss, and an estimate of the conductor operating temperature, for DC, single-phase and three-phase AC, from resistivity and cross-section (with temperature correction), from tabulated R and X per kilometre, or from a mV/A/m figure; parallel conductors per phase; the drop limit is a user input (the regulatory limits live in BS 7671 Appendix 4 and Regulation 525 and are not embedded).

| WHAT IT CALCULATES | RELATION |
|---|---|
| Conductor resistance at its operating temperature, for n conductors in parallel | $R = \rho_T L/(A\,n)$, with $\rho_T = \rho_{20}\,[1 + \alpha\,(T_c - 20)]$ |
| Resistance and reactance from per-length figures instead | $R = R' L/n$; $X = X' L/n$ |
| Voltage drop on a DC circuit | $\Delta V = 2\,I\,R$ |
| Voltage drop on a single-phase AC circuit | $\Delta V = 2\,I\,(R\cos\varphi + X\sin\varphi)$ |
| Voltage drop on a three-phase AC circuit | $\Delta V = \sqrt{3}\,I\,(R\cos\varphi + X\sin\varphi)$ |
| Voltage drop from a tabulated mV/A/m figure, which already carries the 2 or √3 | $\Delta V = (\text{mV/A/m})\,I\,L / 1000$ |
| Power lost in the cable | $P_{loss} = 2\,I^2\,R$ (DC and single-phase); $3\,I^2\,R$ (three-phase) |
| Estimated conductor temperature at this current | $T_{c,est} = T_{amb} + (T_{rated} - T_{amb})(I/I_z)^2$ |

Basis: Voltage drop in consumers' installations[1](525); Voltage drop (mV/A/m tables and the temperature-correction method)[1](Appendix 4, section 6); Resistance, resistivity and temperature coefficient[2](Ch. on DC and AC circuits).

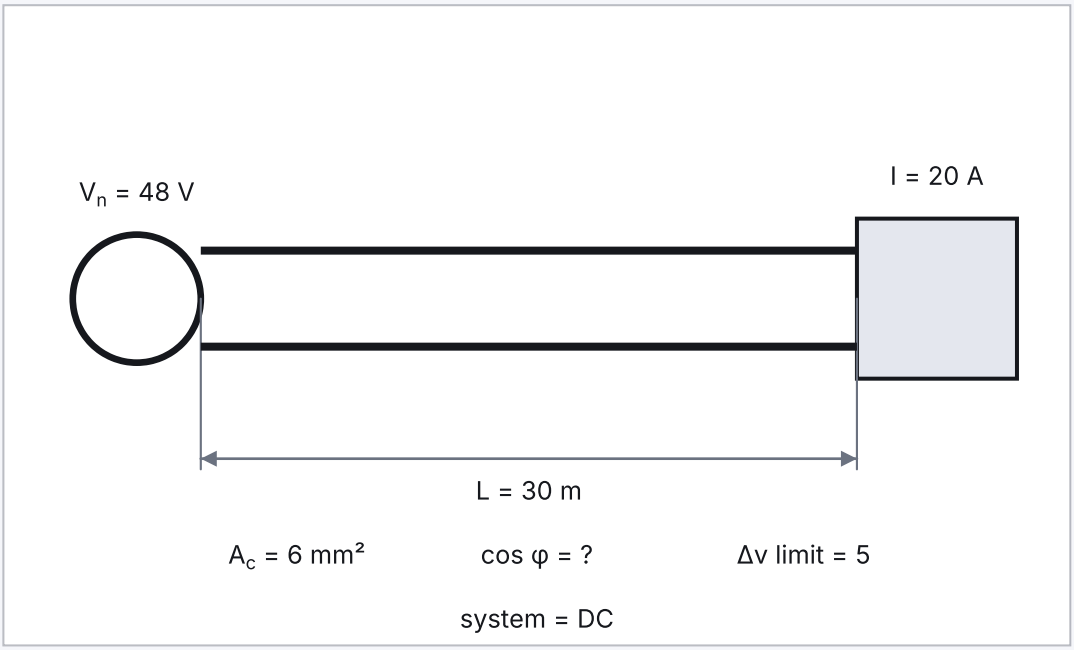



Figure 1 — A DC circuit: the supply, the two conductors of route length L, and the load current I. Not to scale; the labels are the inputs as entered.

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| system | System | DC | | assumption: Design convention. | any |
| $V_n$ | Nominal voltage (line-to-line for three-phase) | 48 | V | assumption: Design convention. | 12 … 1000 V |
| I | Design current | 20 | A | spec: Load schedule DC-02: 20 A | 0.1 … 5000 A |
| L | Route length (one way) | 30 | m | drawing: DRG-E-301 rev A | 0.1 … 5000 m |
| $cos_{phi}$ | Power factor (AC) | 0.85 | | default | 0.5 … 1 |
| $n_{par}$ | Parallel conductors per phase | 1 | | assumption: Design convention. | 1 … 6 |
| data | Conductor data from | resistivity and area | | assumption: Design convention. | any |
| $\rho_{20}$ | Resistivity at 20 °C, the effective value for the conductor resistance (default 0.0194 Ω·mm²/m: the largest class 2 maximum conductor resistance of the cable standard for 1.5 to 120 mm², so the drop is not below the cable tables; pure annealed copper is 0.01724, aluminium 0.0282 Ω·mm²/m) | 0.01724 | ohm·mm²/m | reference: Reference record: annealed copper (Hughes) | 0.01 … 0.05 ohm·mm²/m |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| $\alpha$ | Temperature coefficient of resistance | 0.00393 | | reference: Reference record: annealed copper (Hughes) | 0.003 … 0.0045 |
| $A_c$ | Conductor cross-section | 6 | mm² | spec: Cable schedule CS-02: 6 mm² Cu | 1 … 1000 mm² |
| $T_c$ | Conductor operating temperature for the resistance | 70 | °C | assumption: Design convention. | 20 … 90 °C |
| $X_{km}$ | Reactance per km per conductor | 0 | ohm/km | assumption: Design convention. | 0 … 1 ohm/km |
| $dv_{limit}$ | Voltage-drop limit | 5 | | spec: Battery system spec BS-1: 5 % at the load | 1 … 10 |
| $T_{rated}$ | Conductor rated temperature | 70 | °C | default | 60 … 90 °C |
| $T_{amb}$ | Ambient temperature | 30 | °C | default | 0 … 60 °C |

## 3. Assumptions

A1. Balanced load; conductor reactance from the stated X′ (zero when using resistivity alone unless $X_{km}$ is given); the drop is the arithmetic (R cos φ + X sin φ) form. (Standard engineering approximation.)
A2. A mV/A/m figure is used as tabulated, with its own temperature and system convention. (Cable table convention.)
A3. The conductor temperature estimate scales the rated rise with $(I/I_z)^2$; $I_z$ is the user's tabulated value for the installation method and corrections. (BS 7671 Appendix 4 approach, formula only.)
A4. system (System) is taken as DC. (Design convention.)
A5. $V_n$ (Nominal voltage (line-to-line for three-phase)) is taken as 48 V. (Design convention.)
A6. $n_{par}$ (Parallel conductors per phase) is taken as 1. (Design convention.)
A7. data (Conductor data from) is taken as resistivity and area. (Design convention.)
A8. $T_c$ (Conductor operating temperature for the resistance) is taken as 70 degC. (Design convention.)
A9. $X_{km}$ (Reactance per km per conductor) is taken as 0 ohm/km. (Design convention.)

## 4. Working

[9] $sin_{phi} = 0$
$= 0$
sin φ
[15] $\rho_T = \rho_{20} * (1 + \alpha * (T_c - 20\ degC) / (1\ K))$
$= 0.01724\ ohm·mm²/m \times (1 + 0.00393 \times (70\ degC - 20\ degC) / 1\ K)$
$= 0.020628\ ohm·mm²/m$
Resistivity at the operating temperature
[16] $R = \rho_T * L / (A_c * n_{par})$
$= 0.020628\ ohm·mm²/m \times 30\ m / (6\ mm² \times 1)$
$= 0.10314\ ohm$
Conductor resistance (one conductor, parallel paths combined)
[17] $X = X_{km} * L / n_{par}$
$= 0\ ohm/km \times 30\ m / 1$
$= 0\ ohm$
Conductor reactance

[18] dV = 2 * I * R
= 2 × 20 A × 0.10314 ohm
= 4.1255 V
Voltage drop
[19] $dV_{pct}$ = 100 * dV / $V_n$
= 100 × 4.1255 V / 48 V
= 8.5949
Voltage drop, per cent
[20] $P_{loss}$ = 2 * I^2 * R
= 2 × (20 A)^2 × 0.10314 ohm
= 82.511 W
Conductor power loss
[21] $u_{dv}$ = $dV_{pct}$ / $dv_{limit}$
= 8.5949 / 5
= 1.719
Drop utilisation against the limit

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| R | 0.10314 ohm | Resistance |
| X | 0 ohm | Reactance |
| dV | 4.1255 V | Voltage drop |
| $dV_{pct}$ | 8.5949 | Voltage drop (%) |
| $P_{loss}$ | 82.511 W | Power loss |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| FAIL | Voltage drop ≤ limit | $u_{dv}$ = 1.719 | $u_{dv}$ ≤ 1 | 1.72 | [1](525) |

## 7. Applicability

WARNING — input.cos_phi: $cos_{phi}$ (Power factor (AC)) was not entered; the template default 0.85 is used.
WARNING — input.T_rated: $T_{rated}$ (Conductor rated temperature) was not entered; the template default 70 degC is used.
WARNING — input.T_amb: $T_{amb}$ (Ambient temperature) was not entered; the template default 30 degC is used.

## 8. Verification

Template elec.voltage-drop@1.1.0 Content hash 28e4e44162109057 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

## 9. References

1. British Standards Institution; Institution of Engineering and Technology. Requirements for Electrical Installations: IET Wiring Regulations. BS 7671. IET.
2. Hughes E. Hughes Electrical and Electronic Technology. 12th ed. Pearson; 2016.